

# **UWB Data Collection**

**Innovation and Design Programme  
National University of Singapore**

**31 December 2025**

# **UWB Data Collection**

## **Outline**

- 2DK + 2BP Experiments
- iPhone + 2BP Localization Experiments
- Pose Graph Optimization Experiments

# 2DK + 2BP Experiments

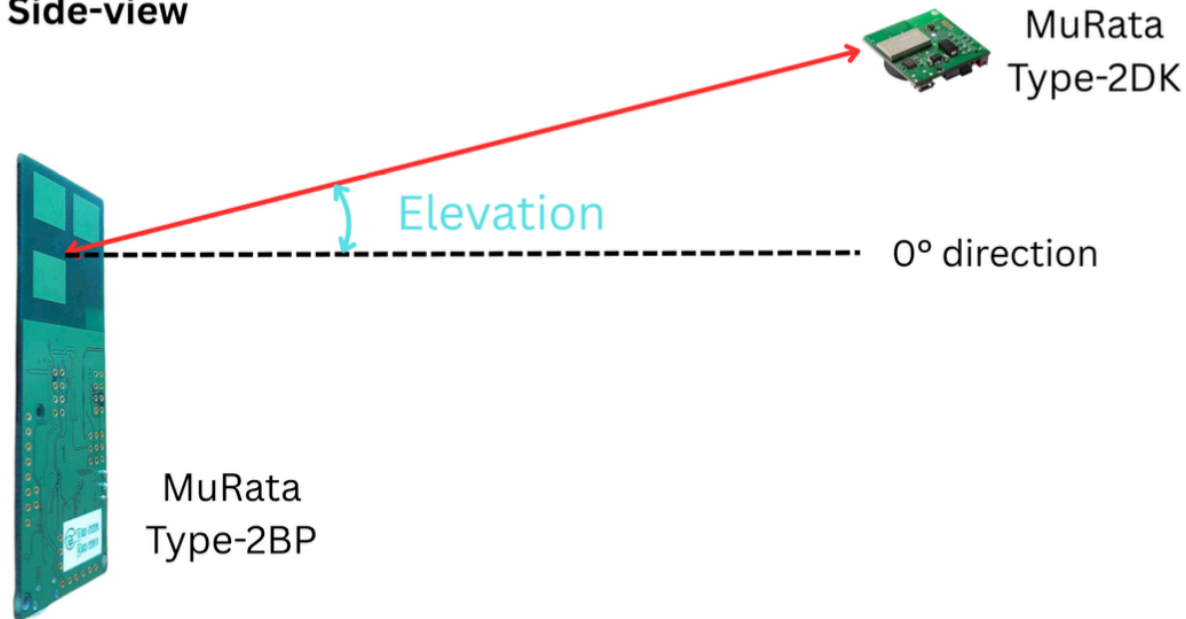
## Experimental Setup

- 1 x Murata 2BP running the standalone app (firmware version: v04.06.05)
- 1 x Murata 2DK (firmware version: v04.03.14)
- With other firmware versions, the 2BP would report the same azimuth value despite the true value being varied
- In addition, values reported by the 2BP did not update fast enough with respect to changes in the location of the 2DK

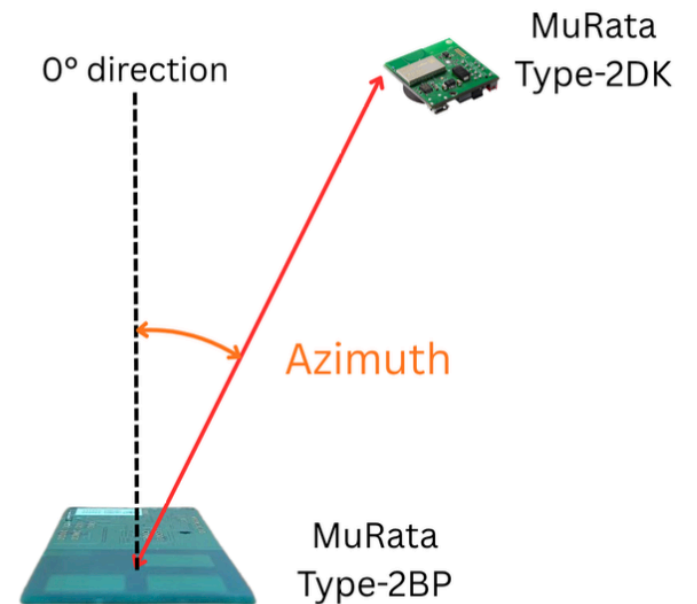
# 2DK + 2BP Experiments

## Experimental Setup

Side-view



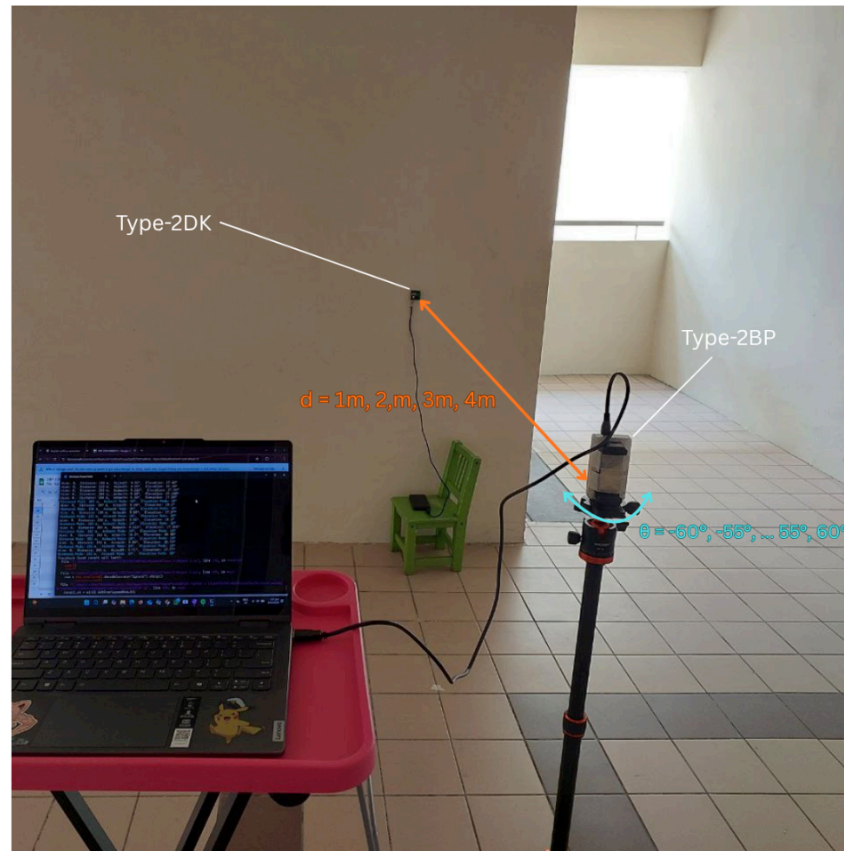
Top-view





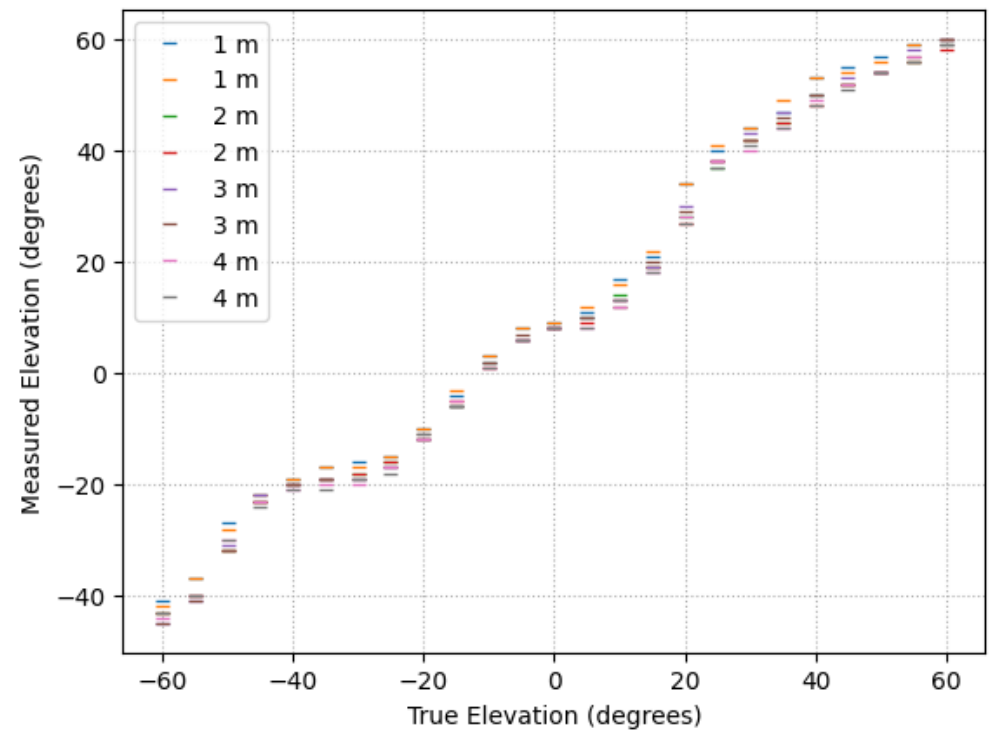
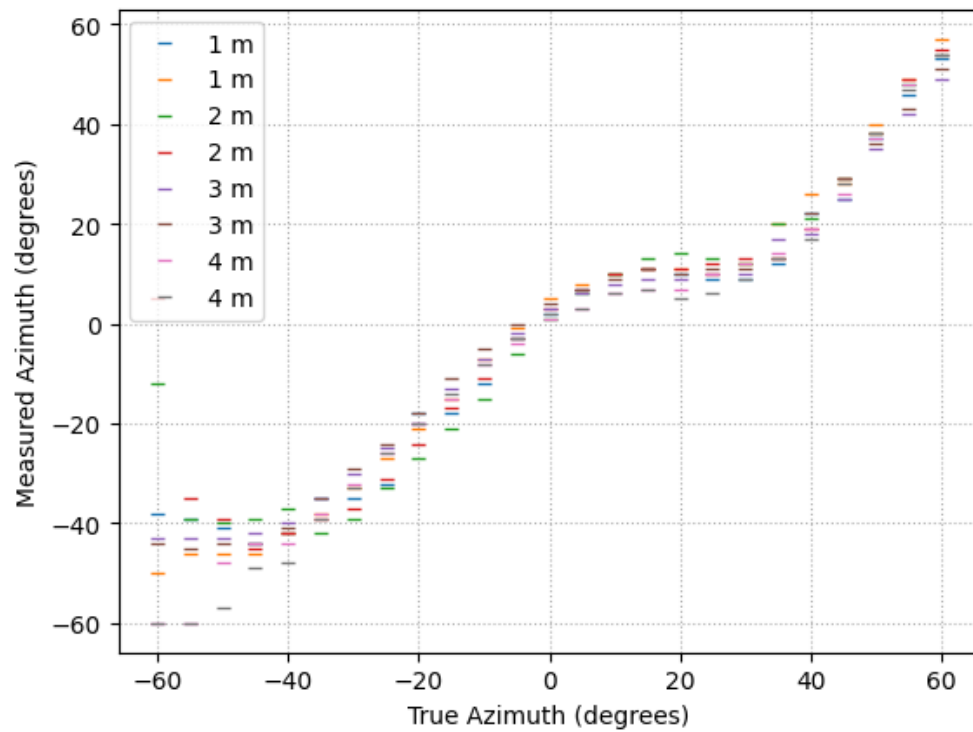
# 2DK + 2BP Experiments

## Experimental Setup



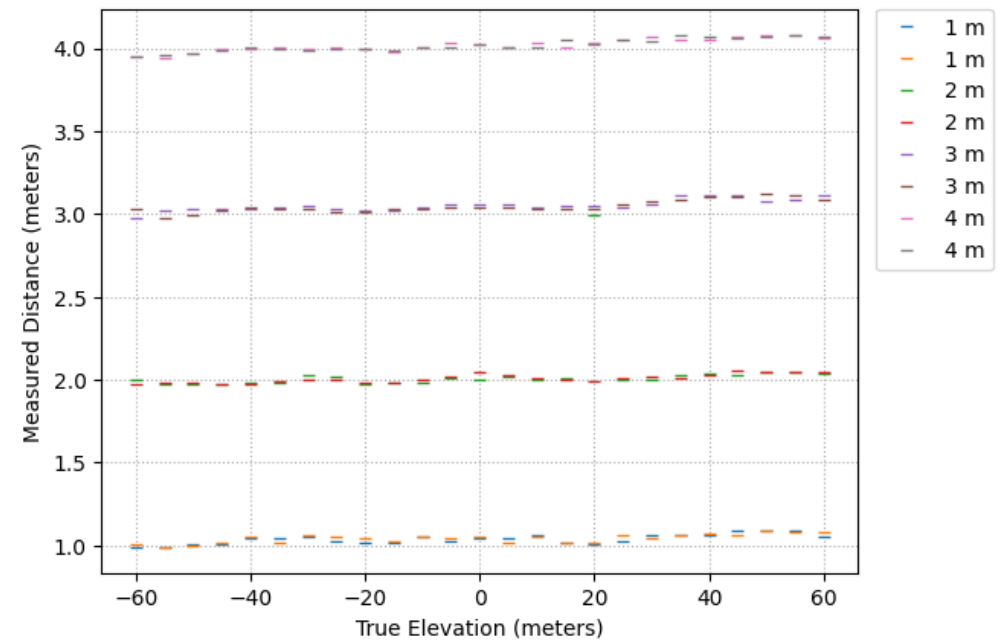
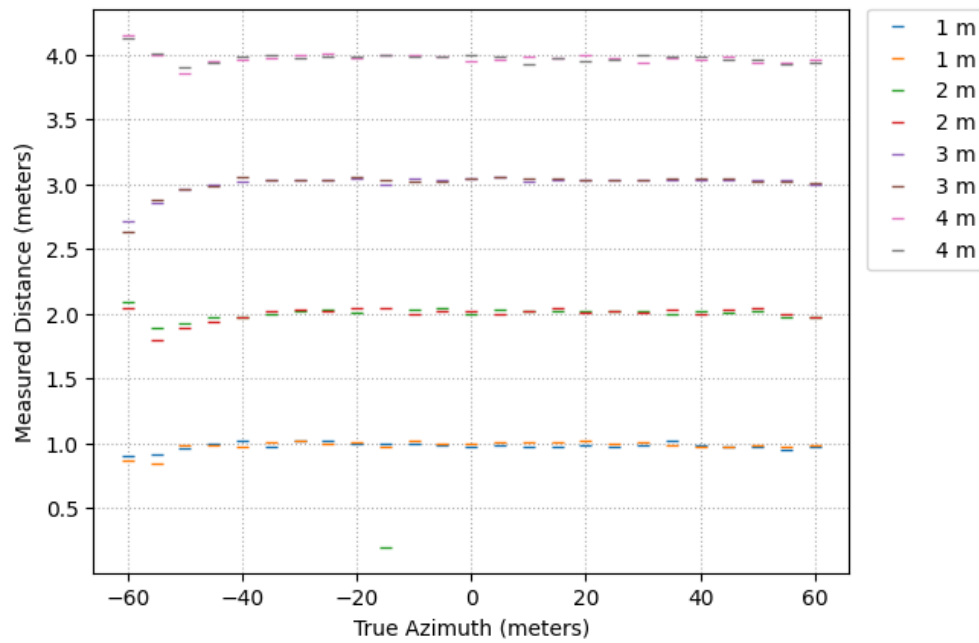
# 2DK + 2BP Experiments

## Results



# 2DK + 2BP Experiments

## Results



# **UWB Data Collection**

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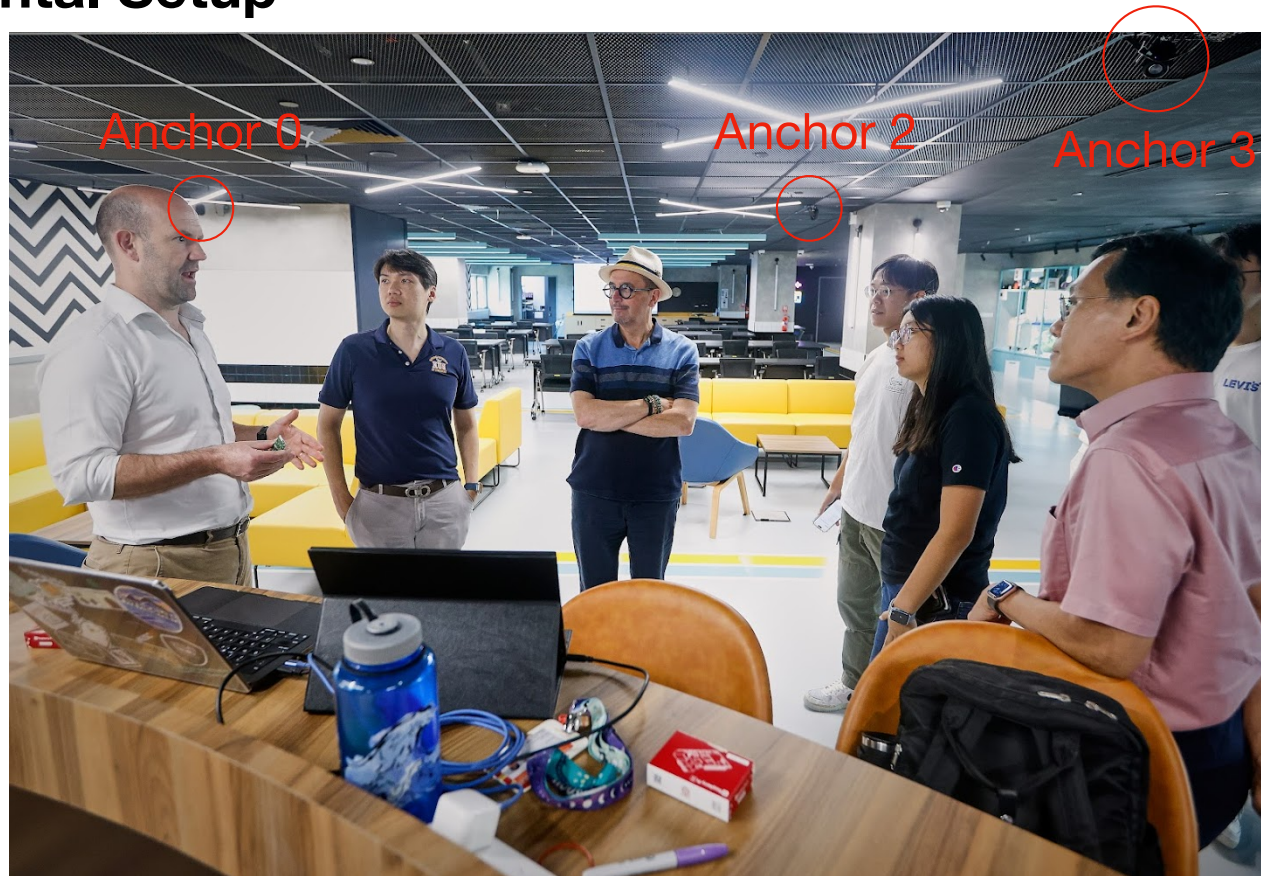
# **iPhone + 2BP Localization Experiments**

## **Experimental Setup**

- 4 x Murata 2BP's running the standalone app (firmware version: v04.06.00)
- Each connected to a Raspberry Pi 4 via USB
- Each Raspberry Pi publishes the UWB measurements via MQTT and serves as 1 of 4 anchors
- 1 iPhone 14 (with UWB support) running NXP's Trimension app serves as the UWB transmitter
- 1 laptop runs the MQTT broker and the localization server

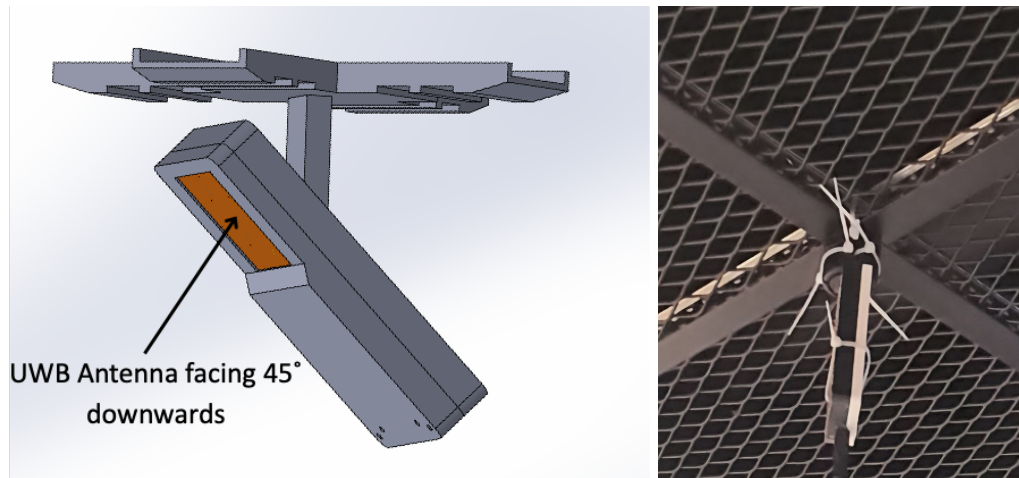
# iPhone + 2BP Localization Experiments

## Experimental Setup



# iPhone + 2BP Localization Experiments

4 Anchors



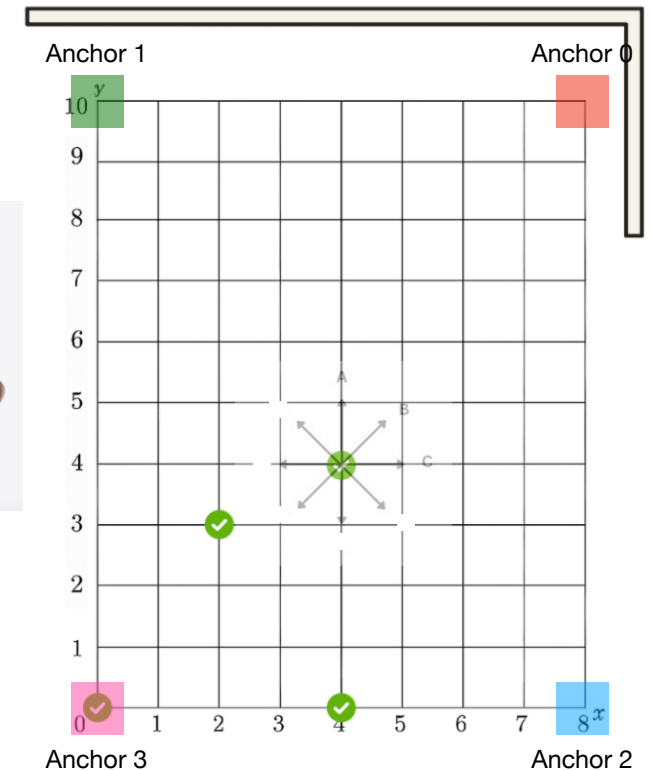
Anchors

4 Orientations

UWB Transmitter  
(iPhone 14)  
Orientation A



4th Orientation  
(Orientation U)  
has the screen of  
the phone facing  
the floor



# iPhone + 2BP Localization Experiments

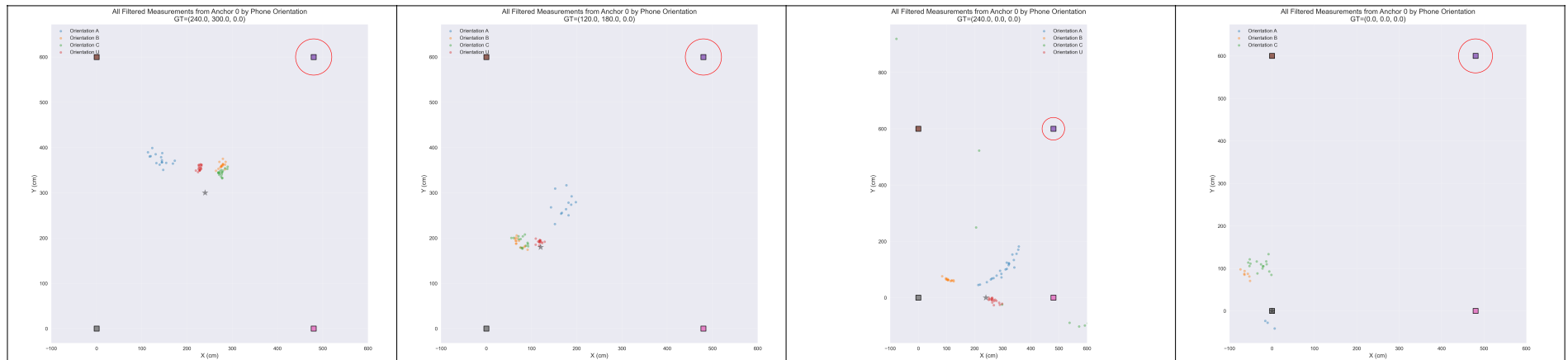
## Data Filtering to Reduce Noise

- In order to reduce noise, readings were averaged over sliding windows of 2 seconds.
- Outliers were removed if their z-scores  $> 2$ .
- At some positions, one of the anchors performed unusually poorly. When the variance of an anchor was too high, we disabled it from inputting into the PGO.
- After implementing this filter, we rejected 10.05% of the data on the basis that they were statistical outliers.



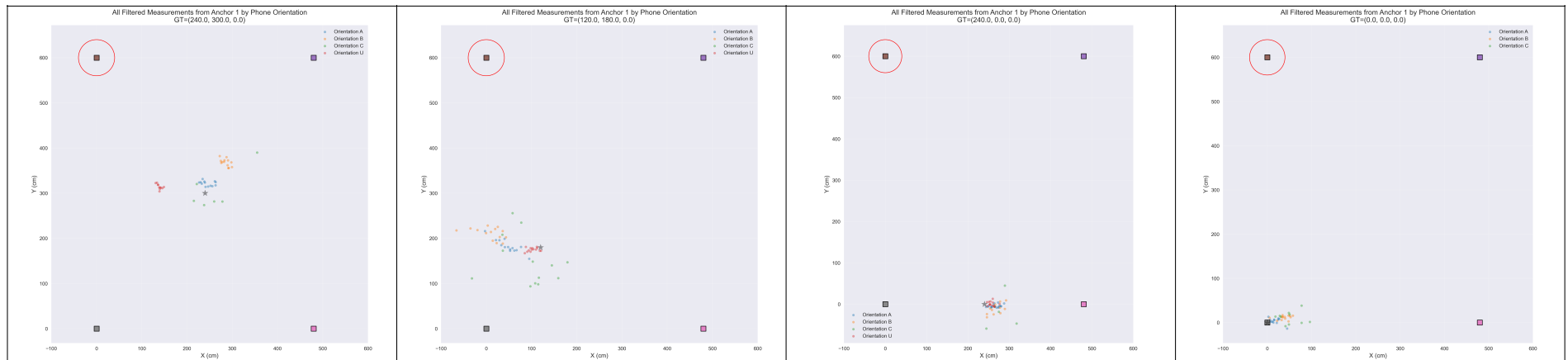
# Anchor 0

## Readings from 4 Ground Truth Positions



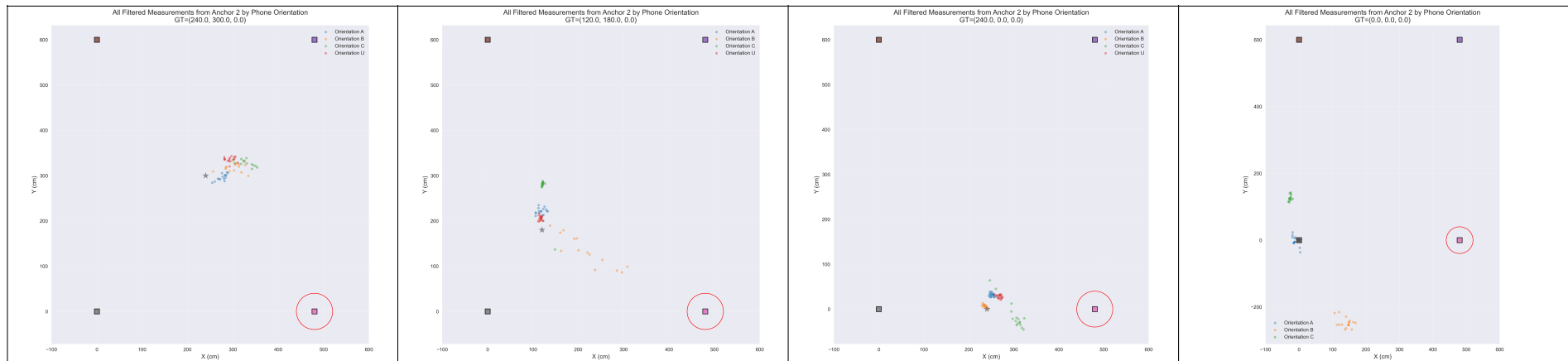
# Anchor 1

## Readings from 4 Ground Truth Positions



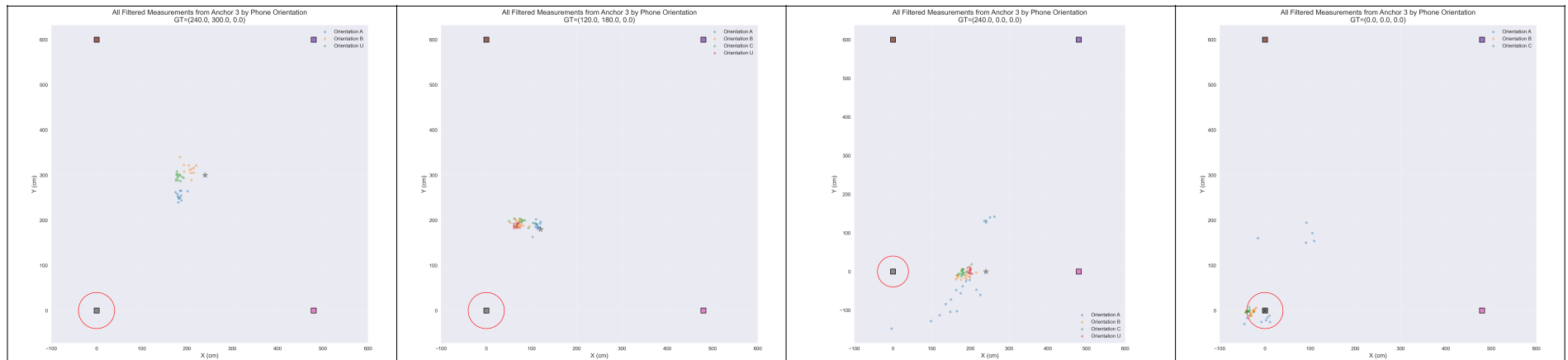
# Anchor 2

## Readings from 4 Ground Truth Positions



# Anchor 3

## Readings from 4 Ground Truth Positions



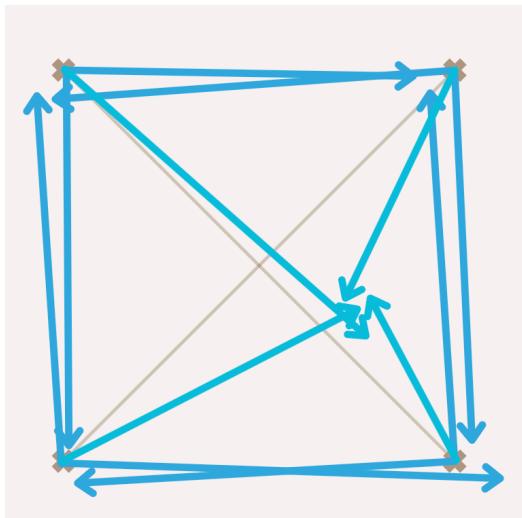
# **UWB Data Collection**

## **Outline**

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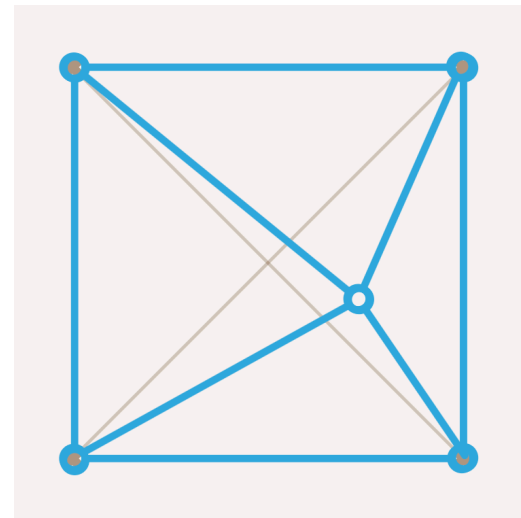
# Pose Graph Optimization Experiments

## Optimization Algorithm



Noisy vector-based  
UWB measurements

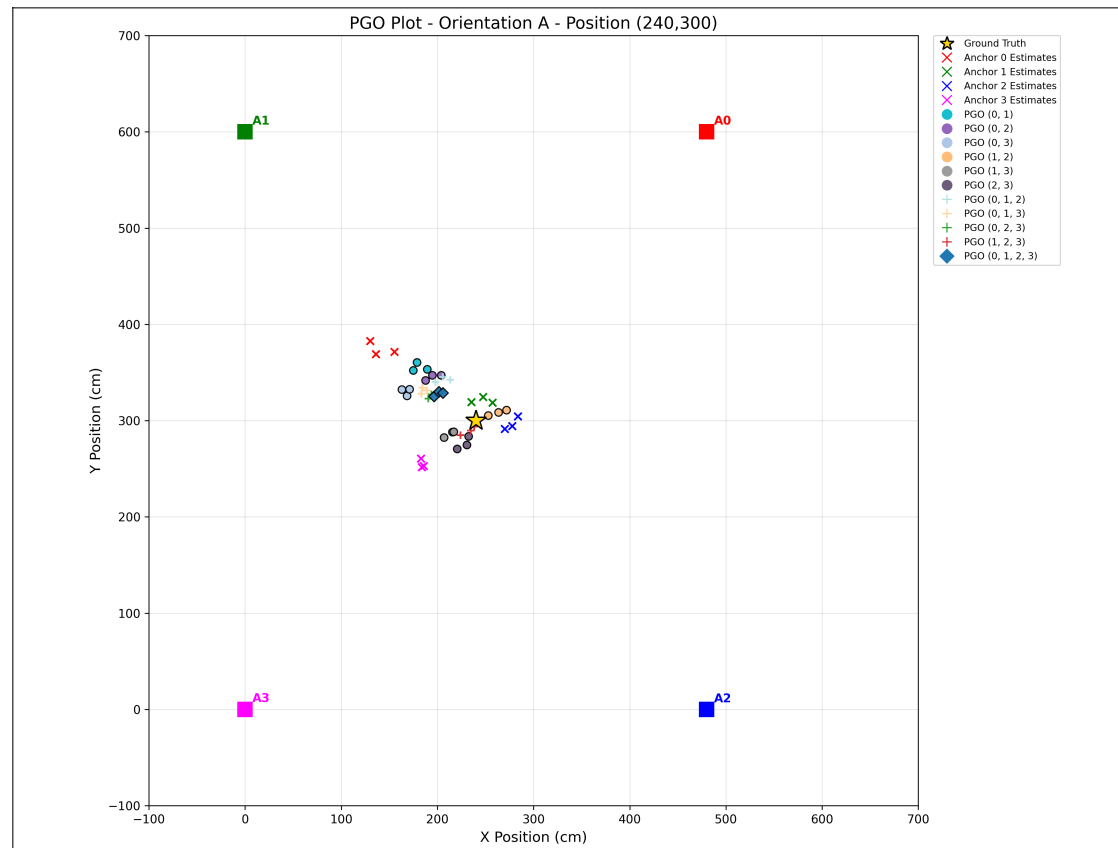
PGO  
Algorithm



Optimized location  
estimate

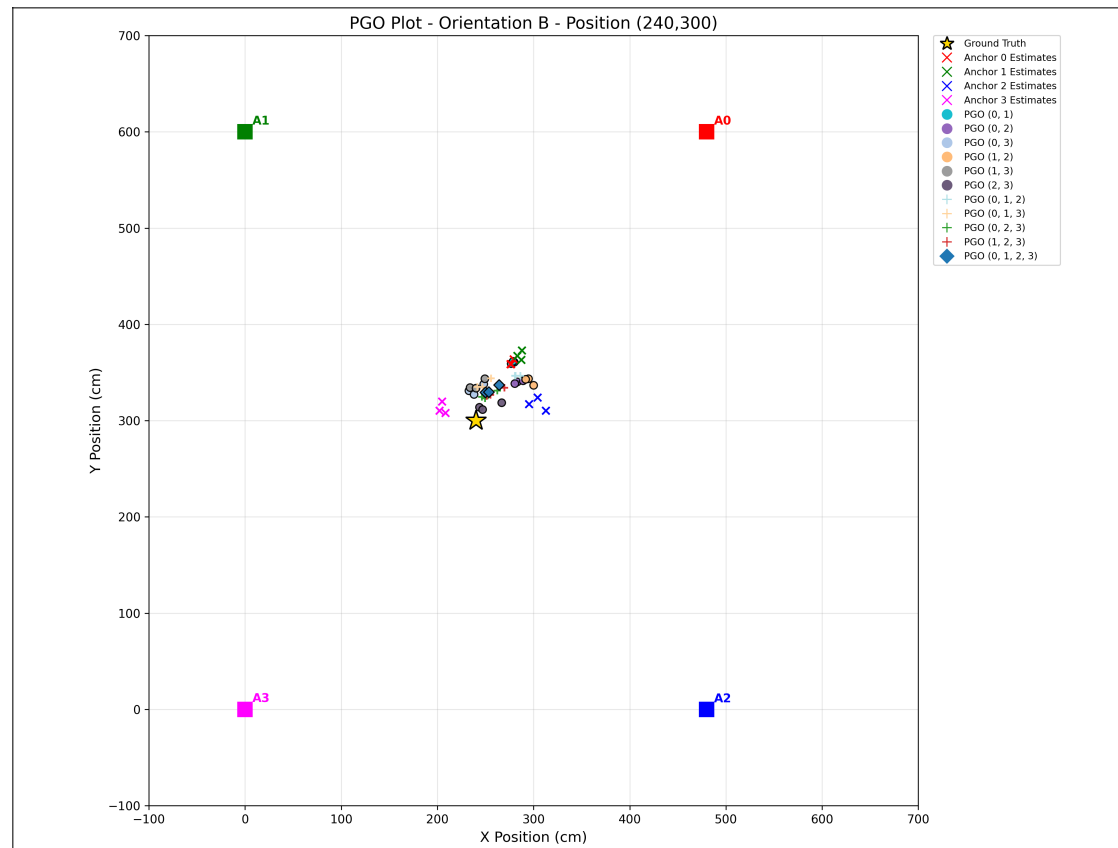
# Pose Graph Optimization

Ground Truth Position (240,300) Orientation A



# Pose Graph Optimization

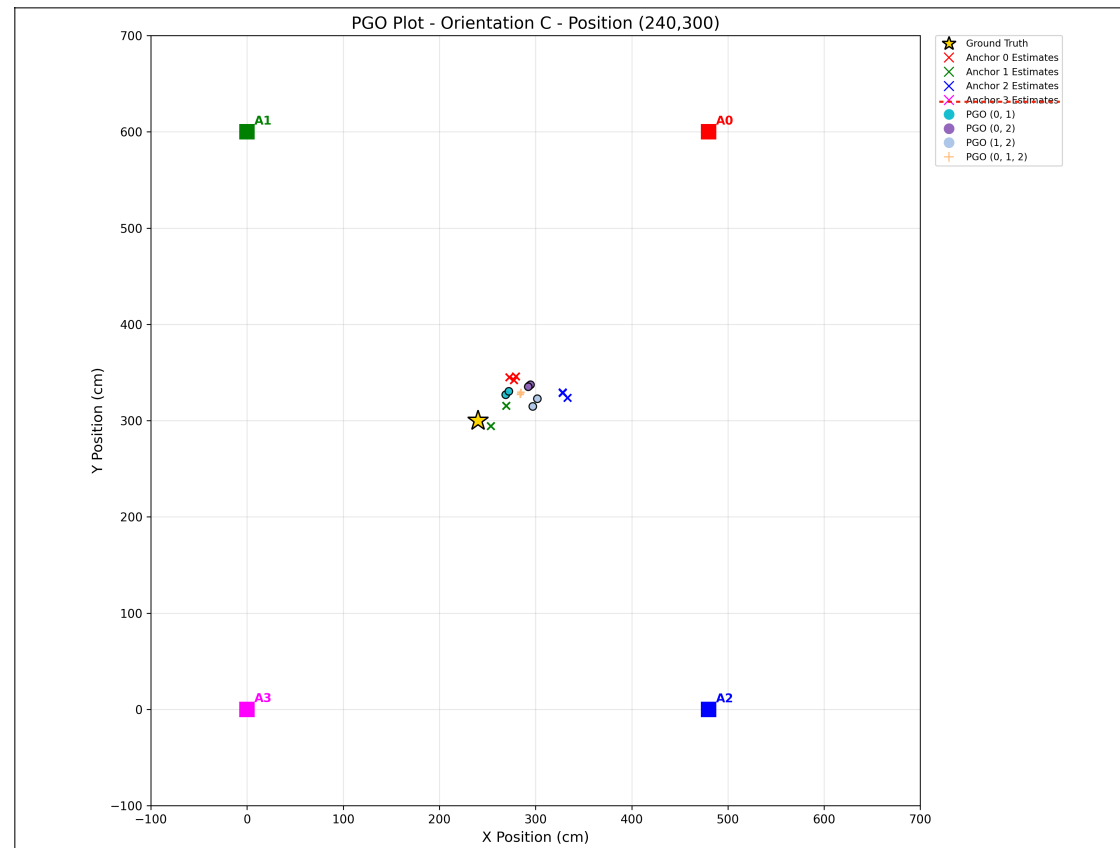
Ground Truth Position (240,300) Orientation B





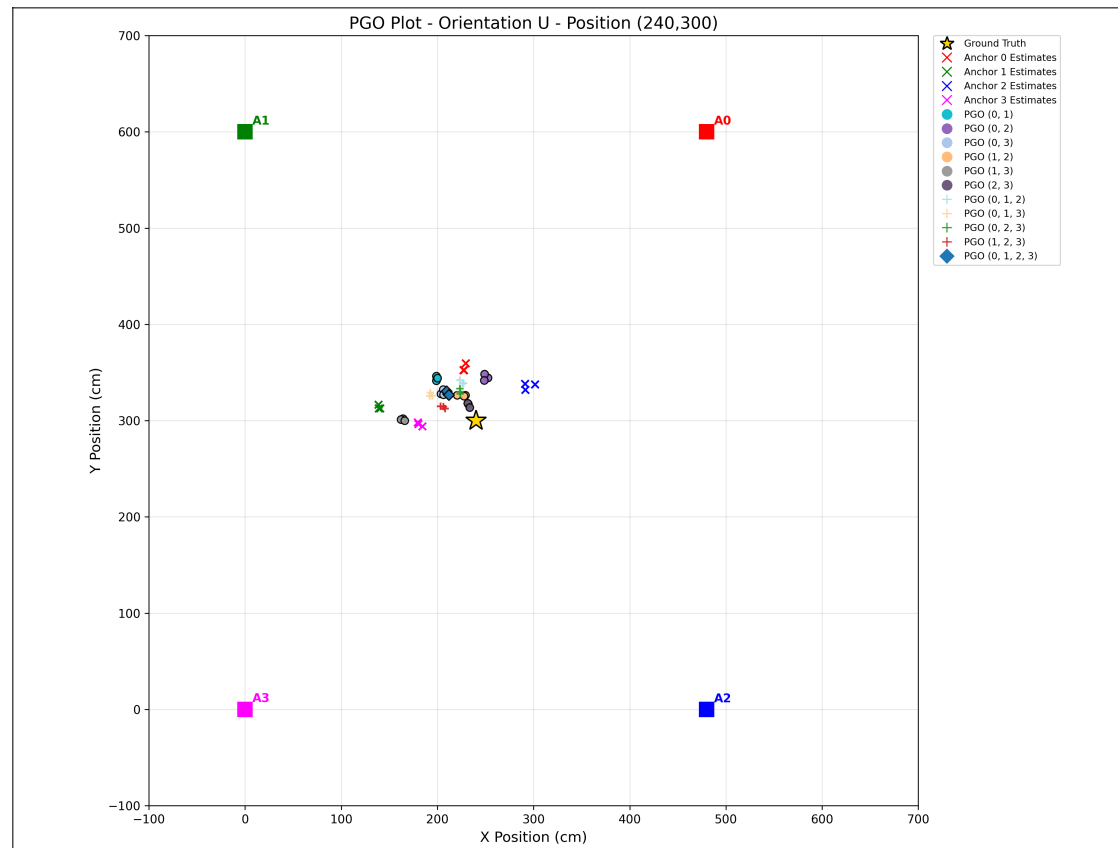
# Pose Graph Optimization

## Ground Truth Position (240,300) Orientation C



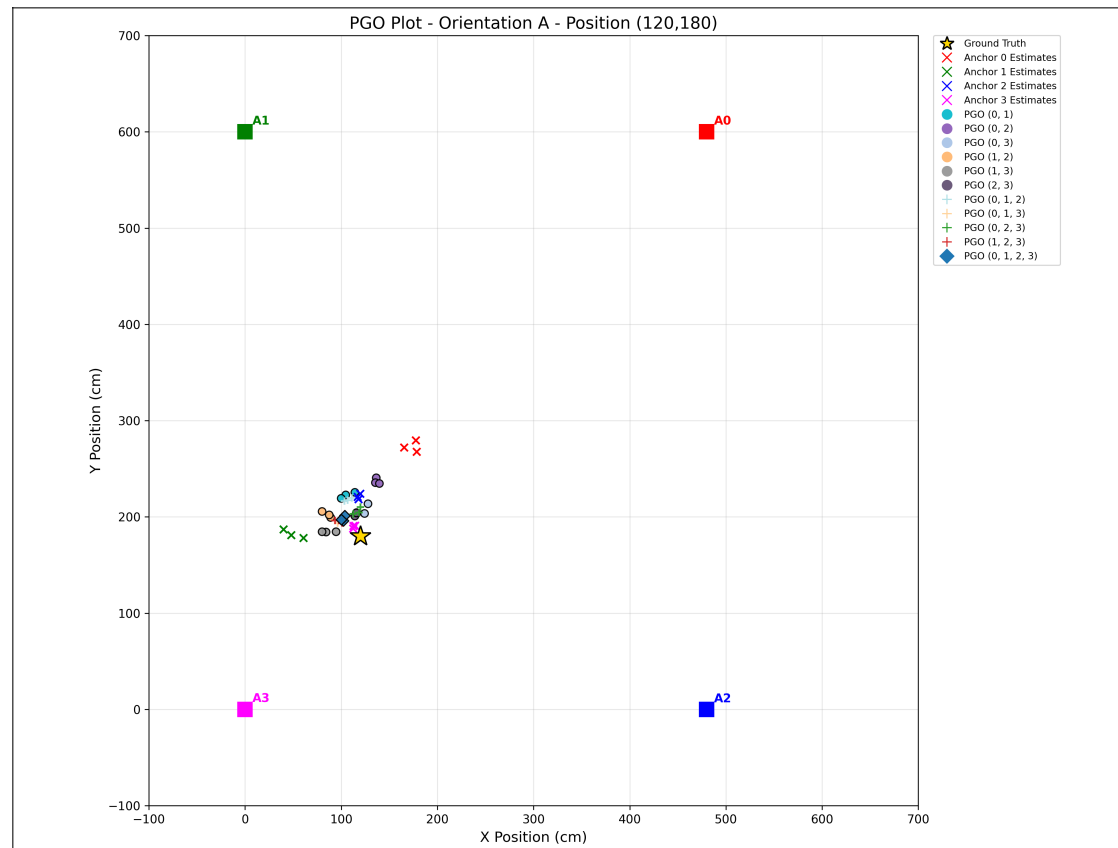
# Pose Graph Optimization

Ground Truth Position (240,300) Orientation U



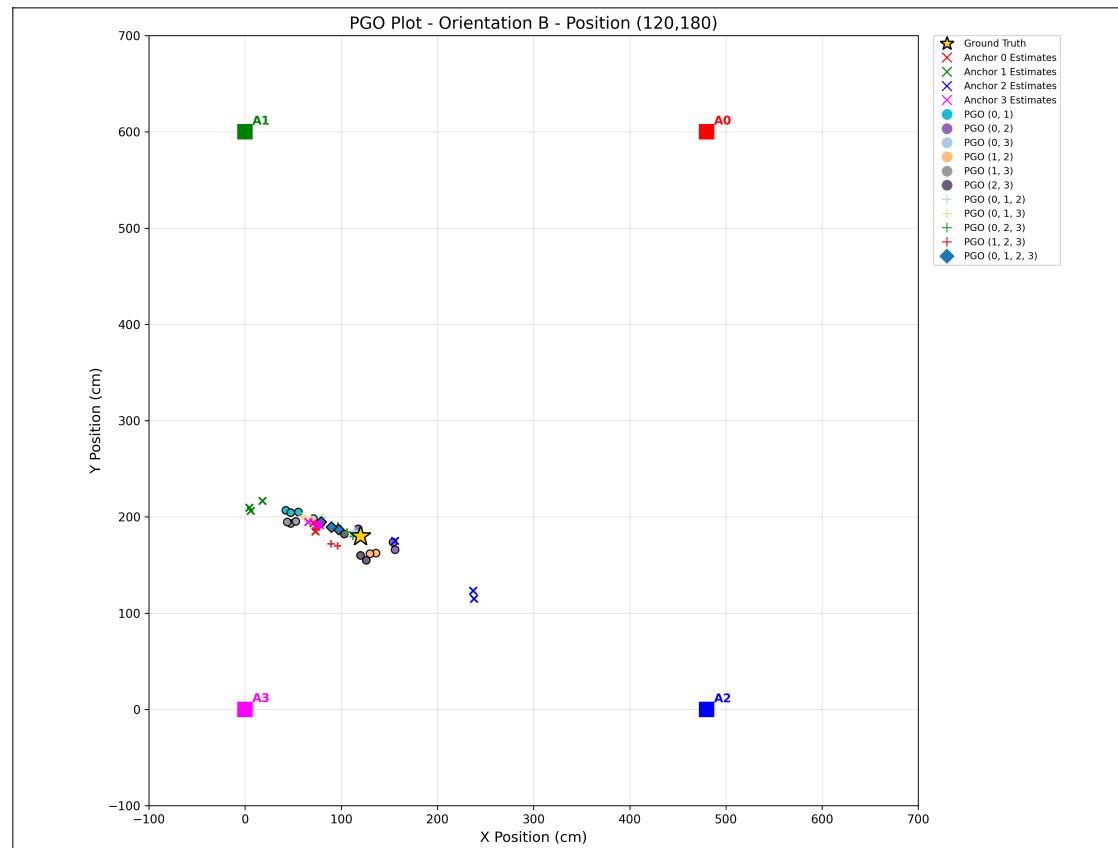
# Pose Graph Optimization

Ground Truth Position (120,180) Orientation A



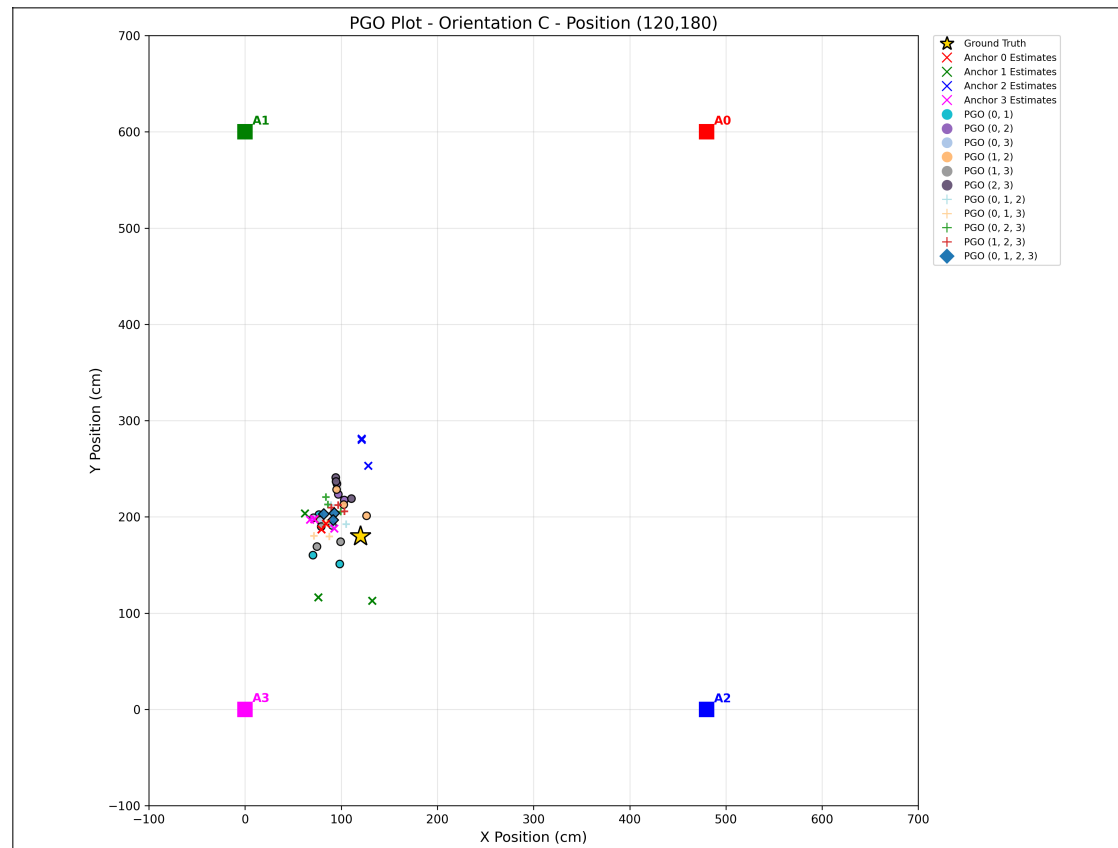
# Pose Graph Optimization

Ground Truth Position (120,180) Orientation B



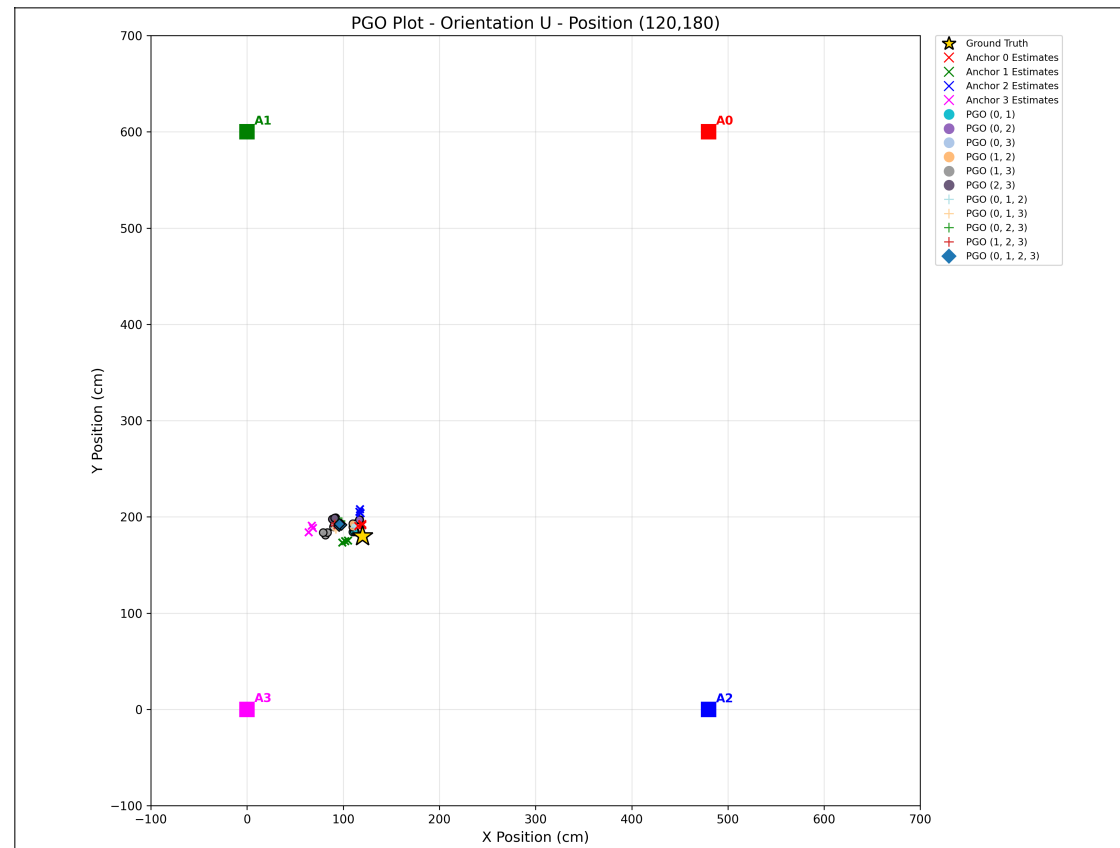
# Pose Graph Optimization

## Ground Truth Position (120,180) Orientation C



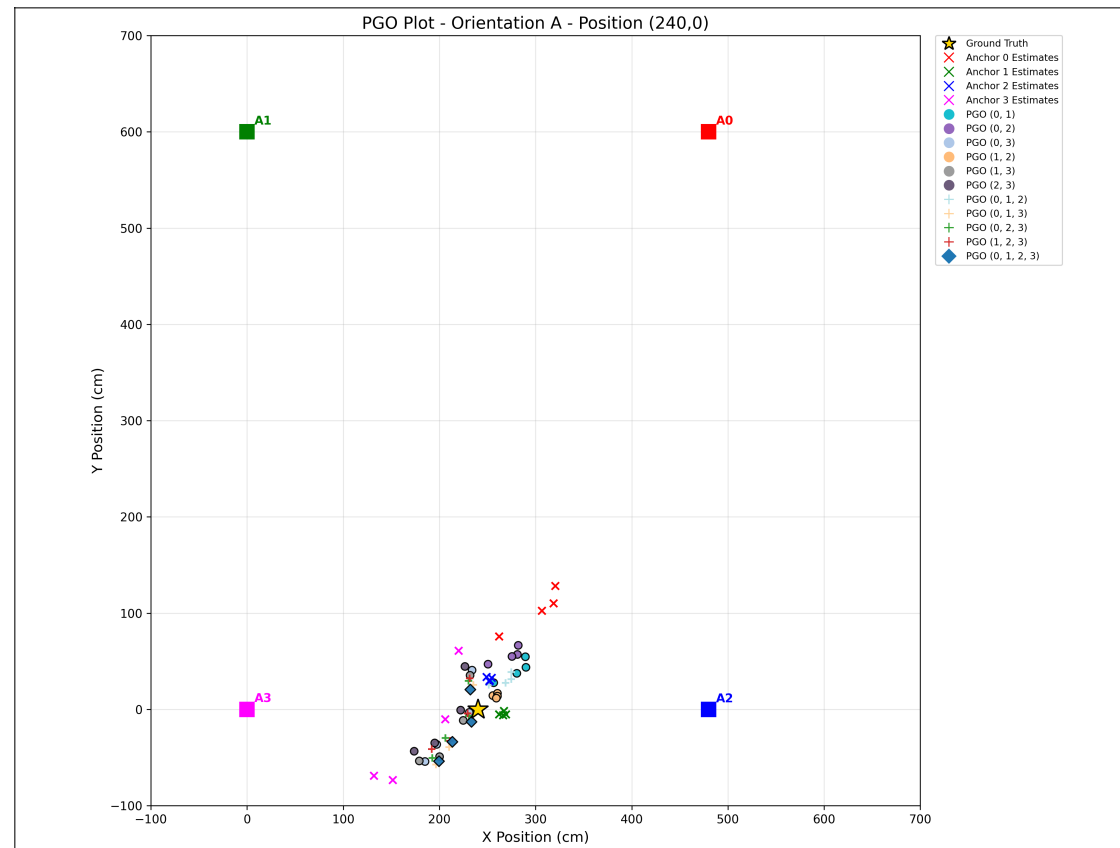
# Pose Graph Optimization

Ground Truth Position (120,180) Orientation U



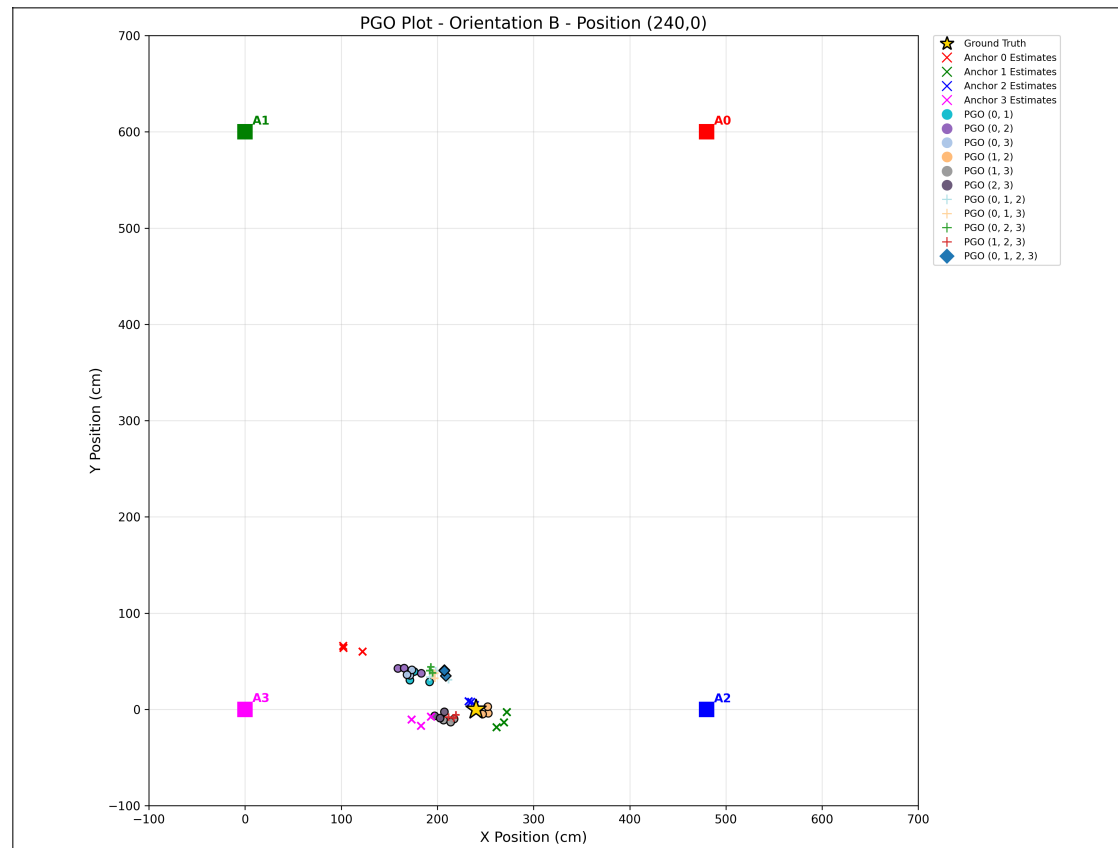
# Pose Graph Optimization

## Ground Truth Position (240,0) Orientation A



# Pose Graph Optimization

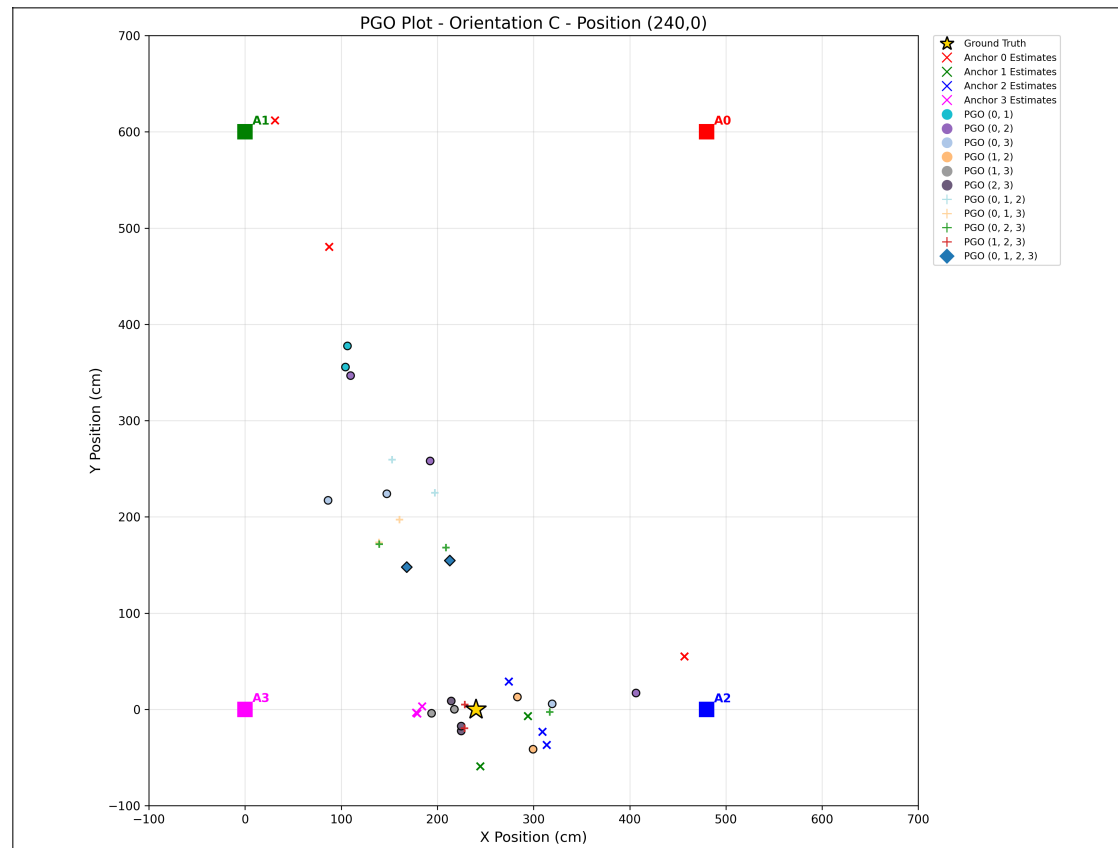
## Ground Truth Position (240,0) Orientation B





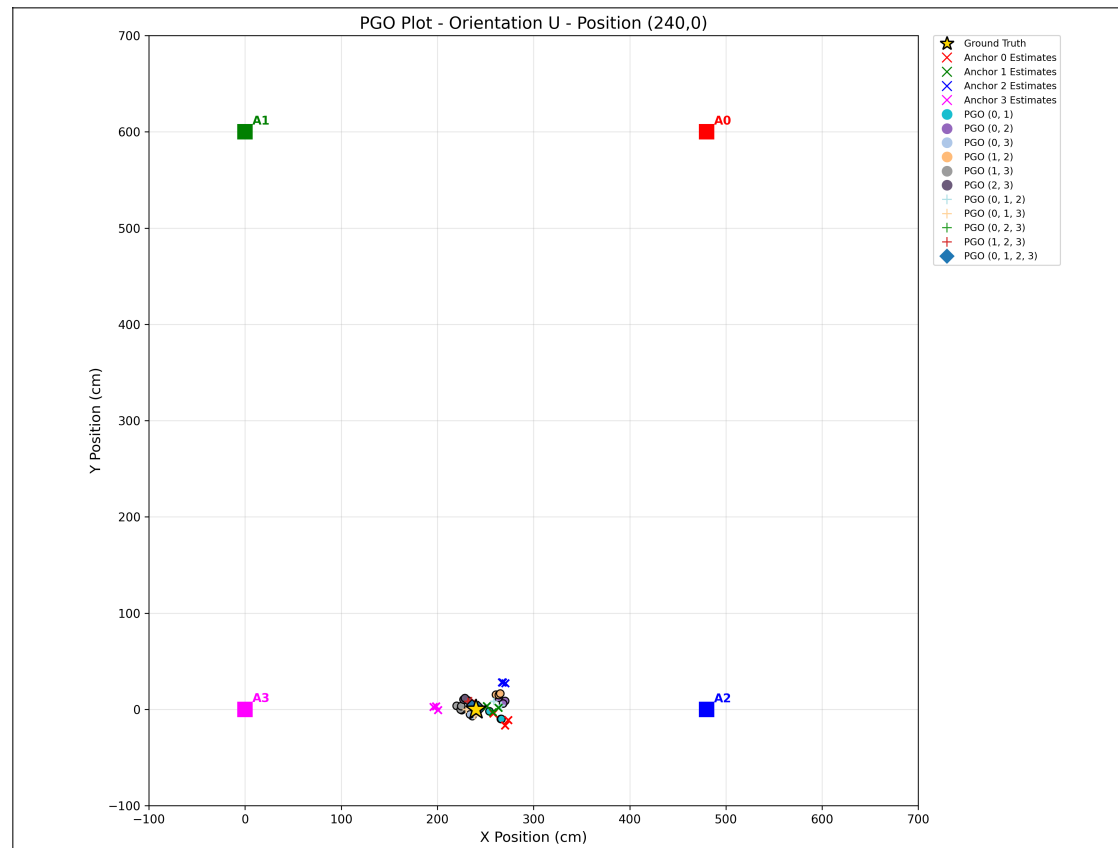
# Pose Graph Optimization

## Ground Truth Position (240,0) Orientation C



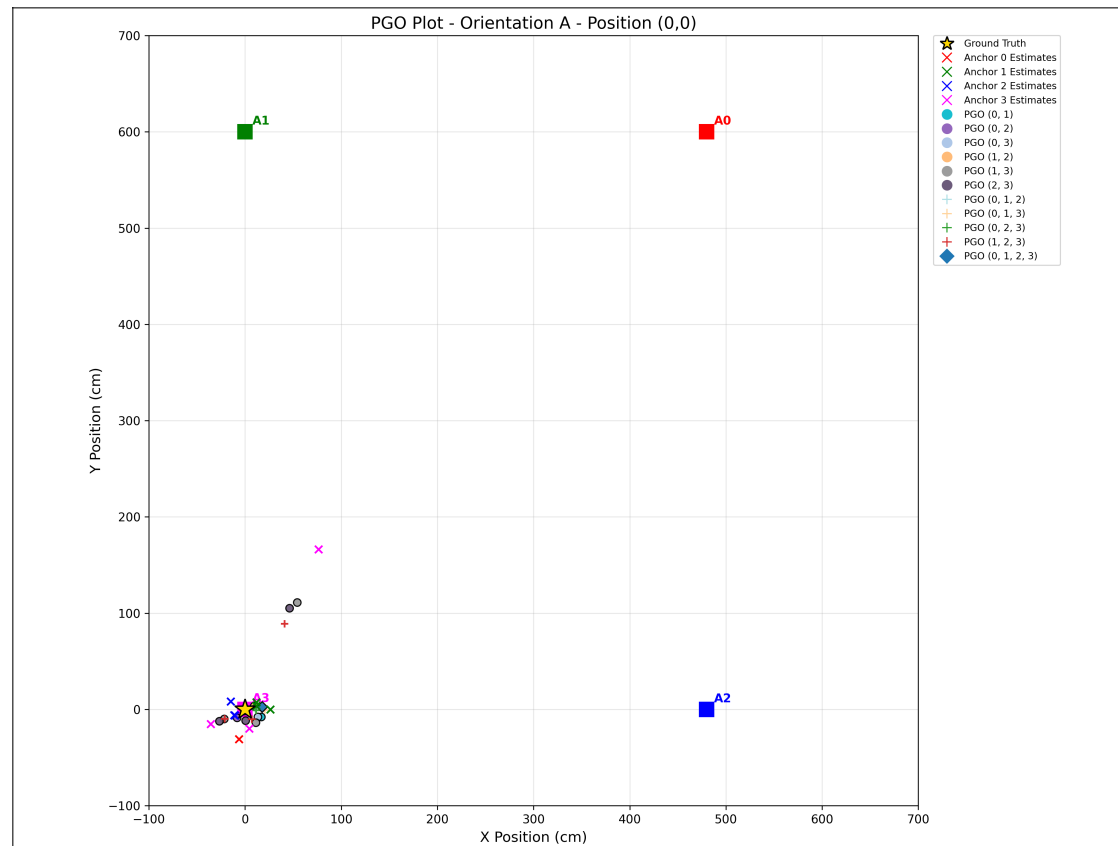
# Pose Graph Optimization

## Ground Truth Position (240,0) Orientation U



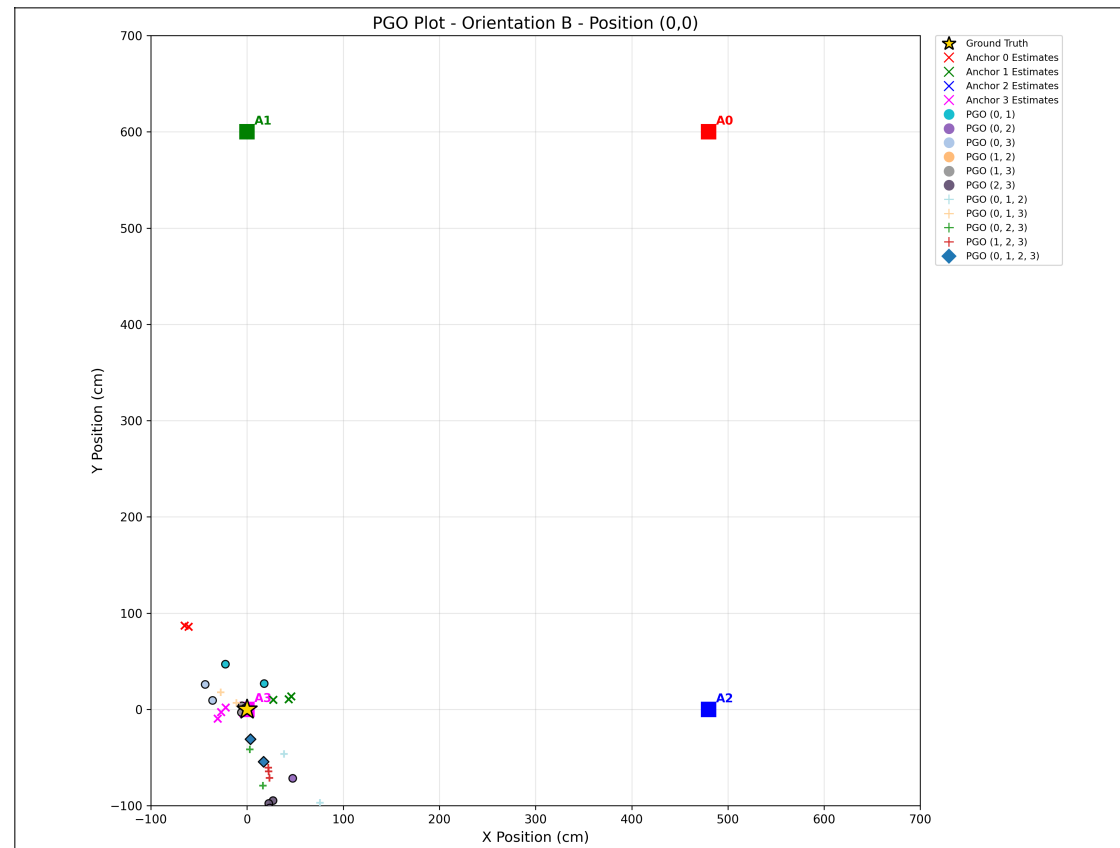
# Pose Graph Optimization

## Ground Truth Position (0,0) Orientation A



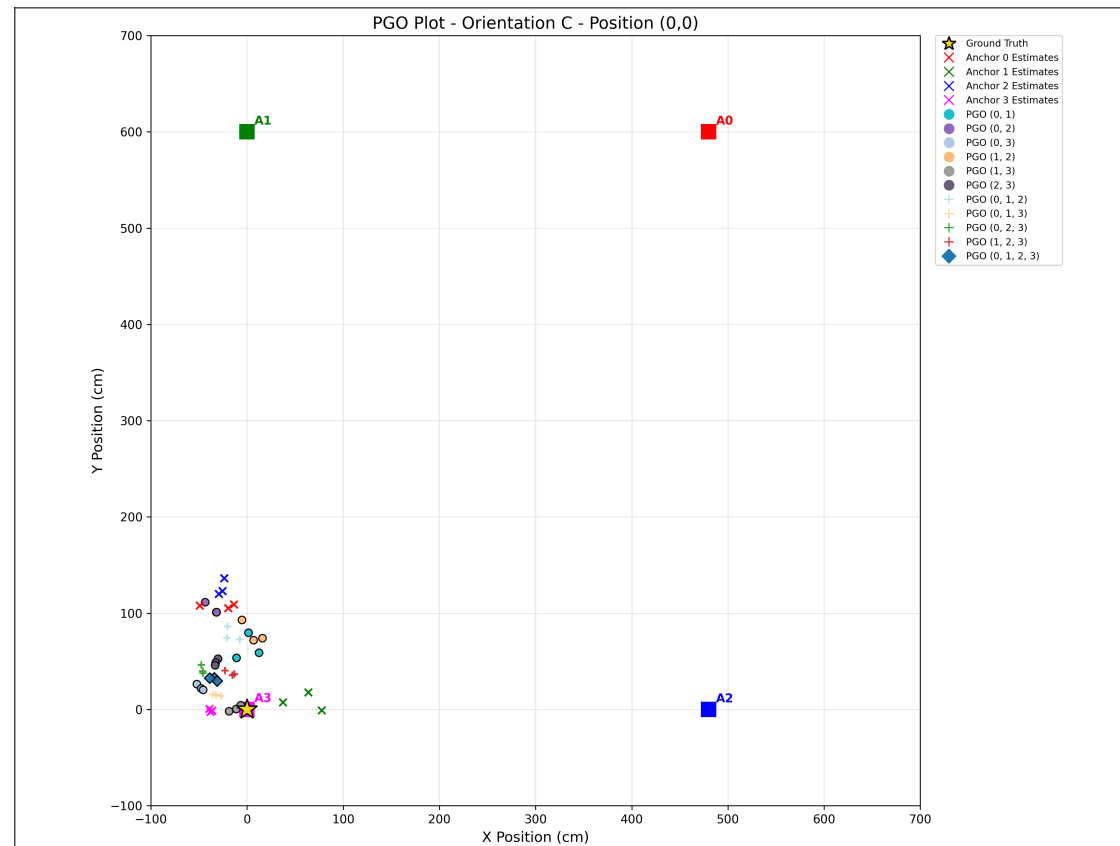
# Pose Graph Optimization

## Ground Truth Position (0,0) Orientation B



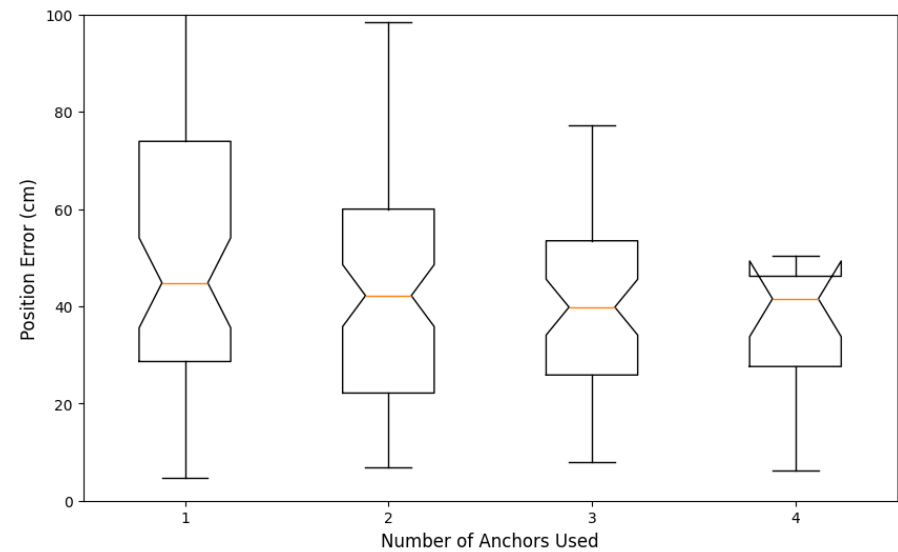
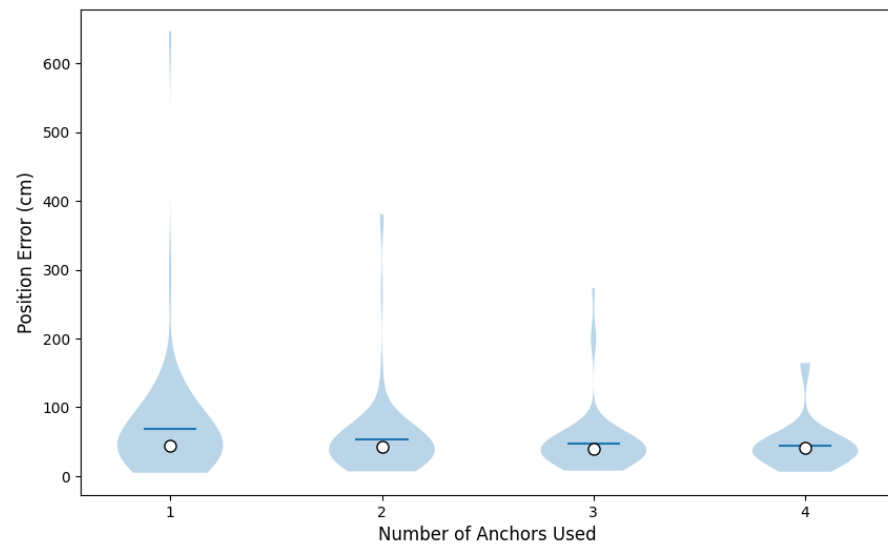
# Pose Graph Optimization

## Ground Truth Position (0,0) Orientation C



# Pose Graph Optimization

## Summary



# Pose Graph Optimization Application Demos



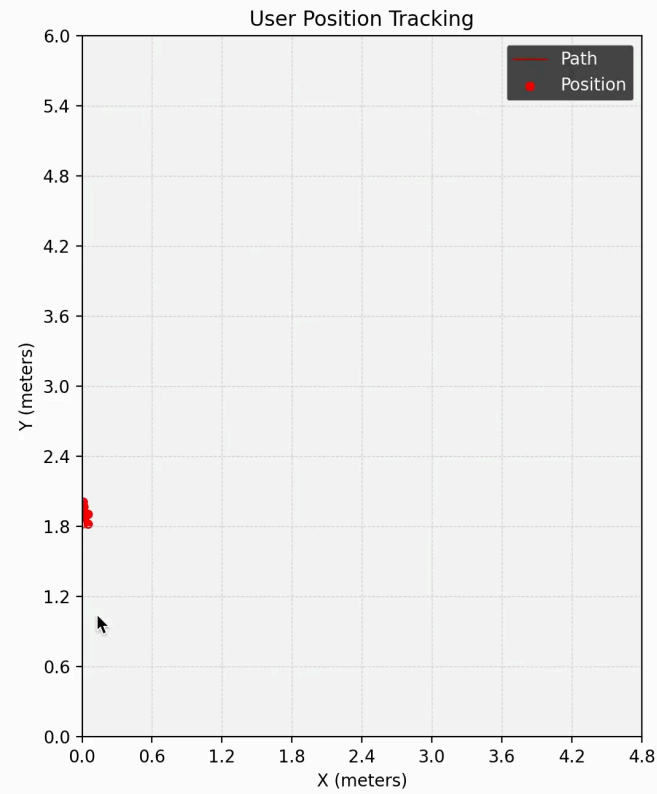
PGO Data

Adaptive Audio

Zone DJ

### PGO Position Data - Real-Time Plot

### Statistics



Pause

Clear

Export CSV

Export PNG

Points: 155 | Paused: No

Pixel: (54, 515) | Meter: (-0.14m, 1.60m) | Server: running





PGO Data

Adaptive Audio

Zone DJ



### Zone DJ

Status

Current Zone: --

Active Zone: --

🎵 crazy-carls-brickhouse-tavern.wav



● Volume

70%

#### Queue Preview

▶ crazy-carls-brickhouse-tavern.wav

✓ funky-bossanova-guitar.wav

✓ guitar-on-loop.mp3

✓ Track 4 - Playlist 1

✓ Track 5 - Playlist 1

# UWB Data Collection

## Summary

- Azimuth readings between a 2BP and a 2DK exhibited low discriminability between the angles of 10 and 35 degrees, as well as -40 and -60 degrees
- Elevation readings exhibited low discriminability between the angles of -5 and 5 degrees, as well as -25 and -40 degrees
- Position readings between a 2BP and an iPhone exhibited errors of approximately 44.8 [37.8, 59.3] cm (median with lower and upper confidence limits)
- Using pose graph optimization, errors using 2 to 4 2BPs changed the errors to: 2 - 42.2 [35.5, 48.3] cm; 3 - 39.8 [32.9, 44.5] cm; 4 - 41.5 [27.5, 45.8] cm